

Project Report: Design and Verification of an 8-Bit SPI Master-Slave Interface

Course Context: VLSI Design

1. Abstract & Introduction

This project details the Verilog RTL development and verification of a Serial Peripheral Interface (SPI) Master-Slave communication protocol. SPI is a synchronous, full-duplex serial communication standard widely used in embedded systems to facilitate high-speed data transfer between a primary microcontroller and peripheral devices, such as OLED displays or sensor modules. The objective of this project was to design the RTL architecture for both the Master and Slave modules, develop a robust testbench, and verify the full-duplex data exchange through waveform simulation using Icarus Verilog.

2. Architecture & The 4-Wire Interface

The SPI protocol operates on a single-master, single-slave (or multi-slave) topology connected via a standard four-wire bus. Unlike asynchronous protocols like UART, SPI relies on a shared clock to guarantee precise timing.

The physical interface consists of the following signals:

- SCLK (Serial Clock): Generated by the Master to synchronize data transmission.
- CS (Chip Select): An active-low signal controlled by the Master to awaken the Slave and initiate a transaction.
- MOSI (Master Out, Slave In): The unidirectional data line carrying payload from the Master to the peripheral.
- MISO (Master In, Slave Out): The unidirectional data line carrying payload from the peripheral back to the Master.

3. Data Transfer Mechanism (Mode 0)

Data transfer in SPI functions fundamentally as a distributed circular shift register. Both the Master and the Slave contain an internal 8-bit register.

This RTL implementation utilizes SPI Mode 0 (Clock Polarity CPOL = 0, Clock Phase CPHA = 0). The mechanics of this mode dictate that:

- The baseline state of the SCLK is 0 (Low) when idle.
- Data is shifted out onto the MOSI and MISO lines on the falling edge of the clock.
- Data is reliably sampled and locked in by the receiving device on the rising edge of the clock.
- A complete transaction requires exactly 8 clock cycles, during which the Master and Slave completely swap their 8-bit register contents simultaneously, achieving true full-duplex communication.

4. RTL Implementation & Conclusion

The Master module was designed using a finite state machine (FSM) comprising two primary states: IDLE and TRANSFER. Upon receiving a start trigger, the Master asserts the CS line low and generates exactly 16 clock edges (8 rising, 8 falling) to facilitate the byte transfer. The Slave module relies on continuous assignment logic to instantly set up the first MISO bit when CS drops, ensuring strict Mode 0 timing compliance.

Simulation waveforms successfully demonstrated that the data vectors (0xA5 and 0x3C) were transmitted and received without any bit-shifting errors or timing violations. The Verilog RTL meets all logical expectations for a synchronous serial interface.